

S-Power

Power, simplified.

Call Centre AI
Data Pipeline

From raw call records to business intelligence

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Portfolio project | November 2026

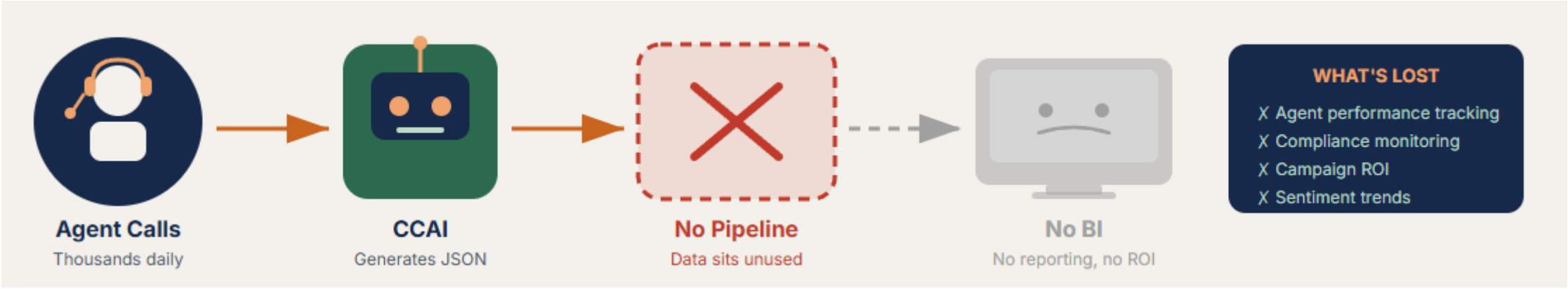
The problem

S-Power is rolling out CCAI

The new system generates thousands of call records daily, each containing speech analytics, sentiment scores, compliance flags and sales outcomes.

No pipeline exists

Without a structured data pipeline, this rich data sits unused. No cleaning, no cloud storage, no reporting, no way to measure ROI or agent performance.



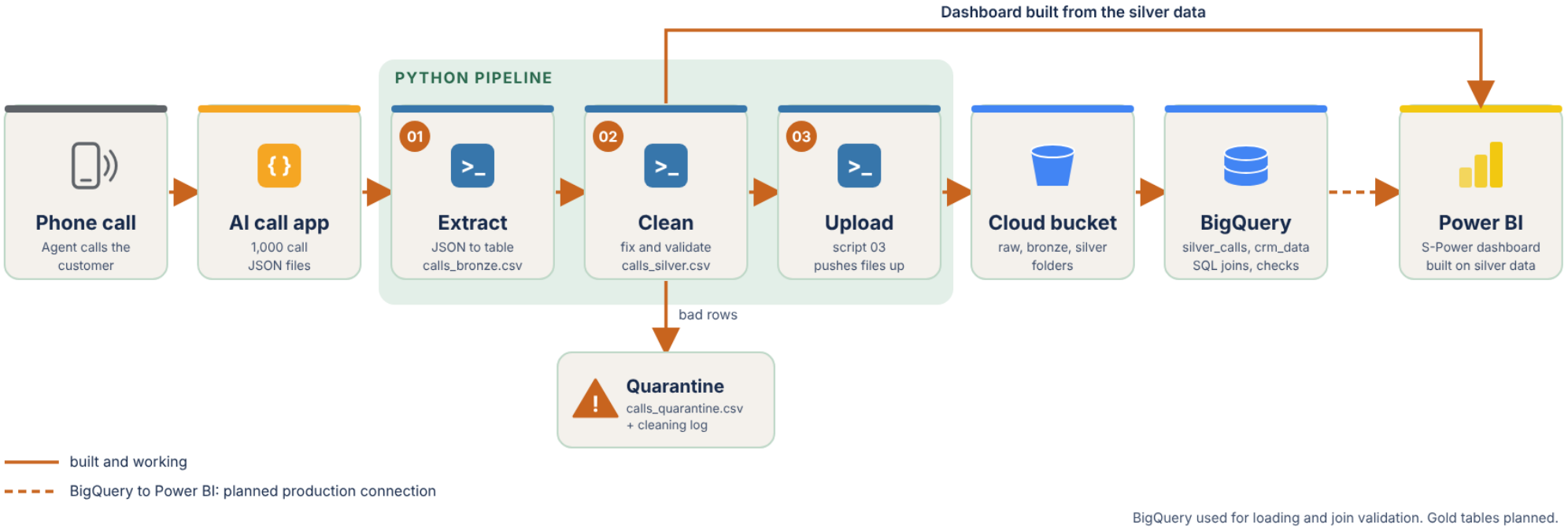
S-Power is a fictional brand. Not affiliated with any real company.

Assumptions

- **Client systems are unknown.** No access to the real telephony platform, CRM, or CCAI configuration. This project simulates what the output would look like.
- **CCAI outputs JSON.** Each call produces a structured JSON record with metadata, transcript summary, sentiment, compliance and sales data.
- **Synthetic data.** 1,000 call records generated with Python to mirror realistic distributions. CRM export simulated separately.
- **Cloud choice is portable.** Built on Google Cloud (GCS + BigQuery) but the medallion pattern works on any cloud platform.

Pipeline overview

Medallion architecture (bronze, silver, gold) with numbered Python scripts per stage. Data flows from raw JSON through cleaning and validation into cloud storage, BigQuery and Power BI.



BigQuery used for loading and join validation. Gold tables planned.

Python pandas, extraction, cleaning, upload

SQL BigQuery joins, view for BI

DAX Power BI measures and KPIs

Bronze: extraction

1,000 synthetic JSON files, each representing one call record from the CCAI system. The extraction script reads every file in the source folder, flattens nested fields, and writes a single bronze-stage CSV.

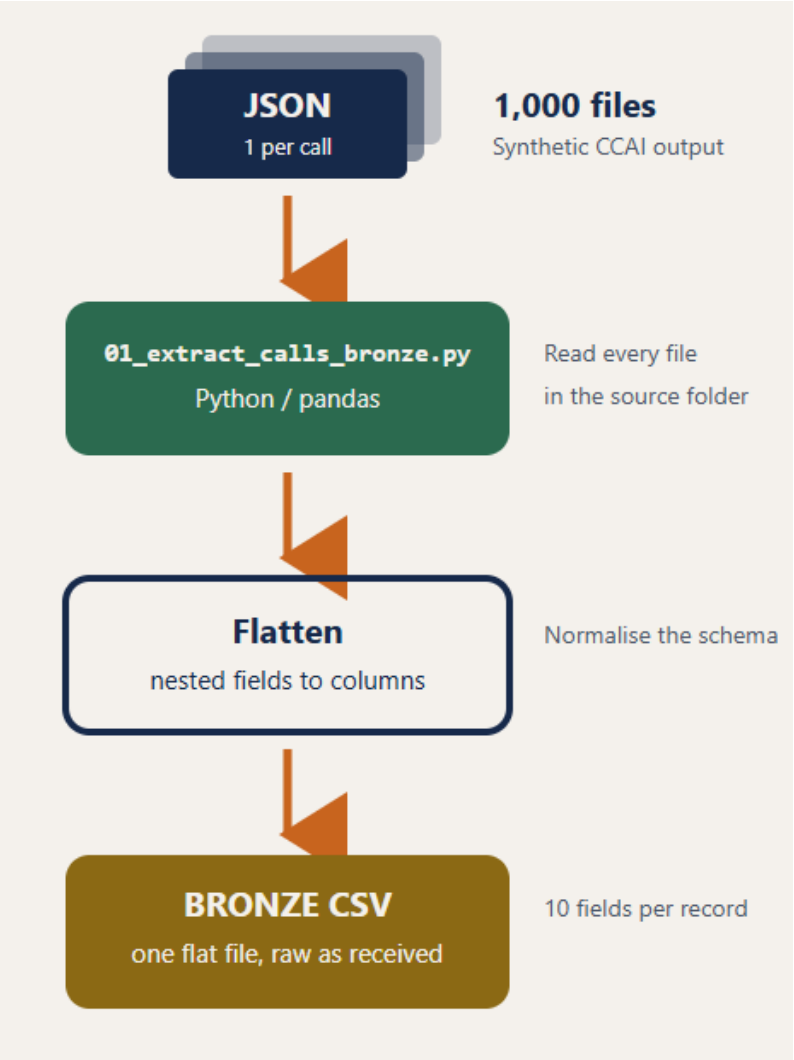
SCRIPT

`01_extract_calls_bronze.py`

Python / pandas: reads JSON, normalises schema, outputs bronze CSV

FIELDS PER RECORD

call_id, agent_id, timestamp, duration, sentiment, compliance_flag, sales_outcome, campaign, intent, transcript_summary



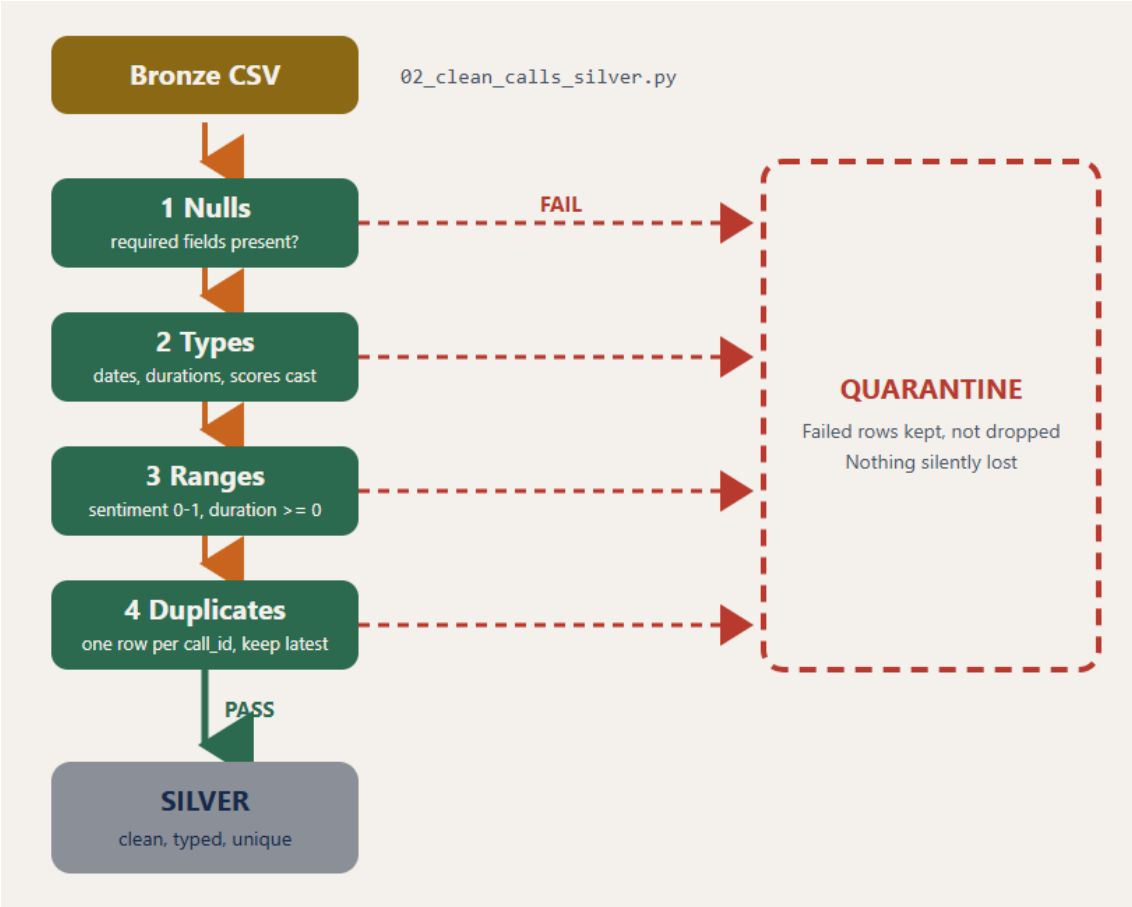
Silver: cleaning

The silver stage validates every row against business rules. Records that fail are quarantined rather than dropped, so nothing is silently lost.

Nulls	Flag and quarantine rows missing required fields
Types	Cast dates, durations and scores to correct types
Ranges	Reject sentiment outside 0-1 or duration below zero
Duplicates	Deduplicate on call_id, keep latest

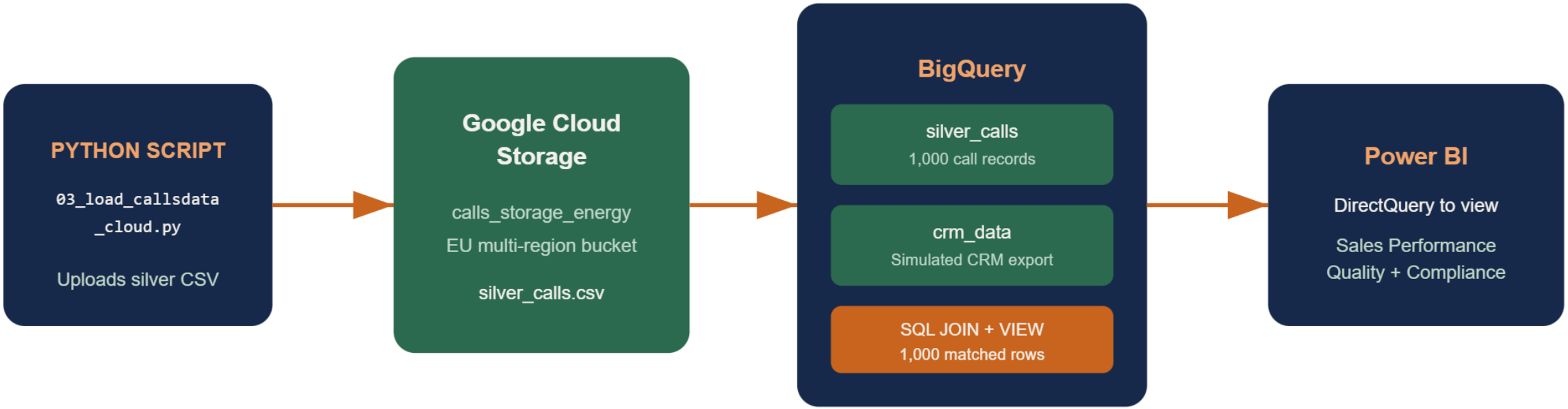
SCRIPT

`02_clean_calls_silver.py`



Cloud infrastructure

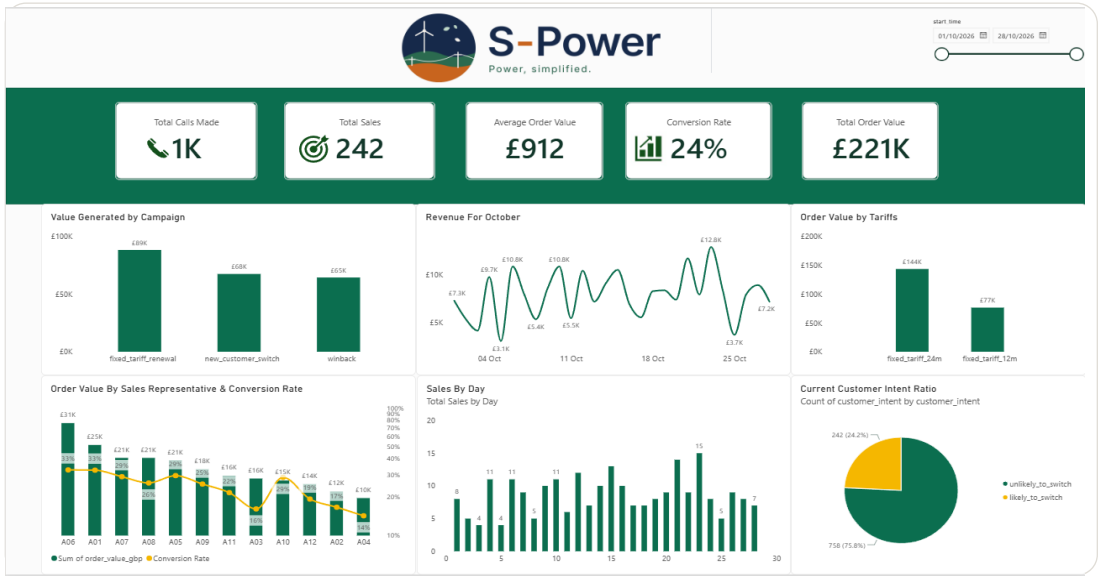
Silver CSV uploaded to Google Cloud Storage, then loaded into BigQuery. SQL joins the call data with a simulated CRM export and creates a unified view for Power BI.



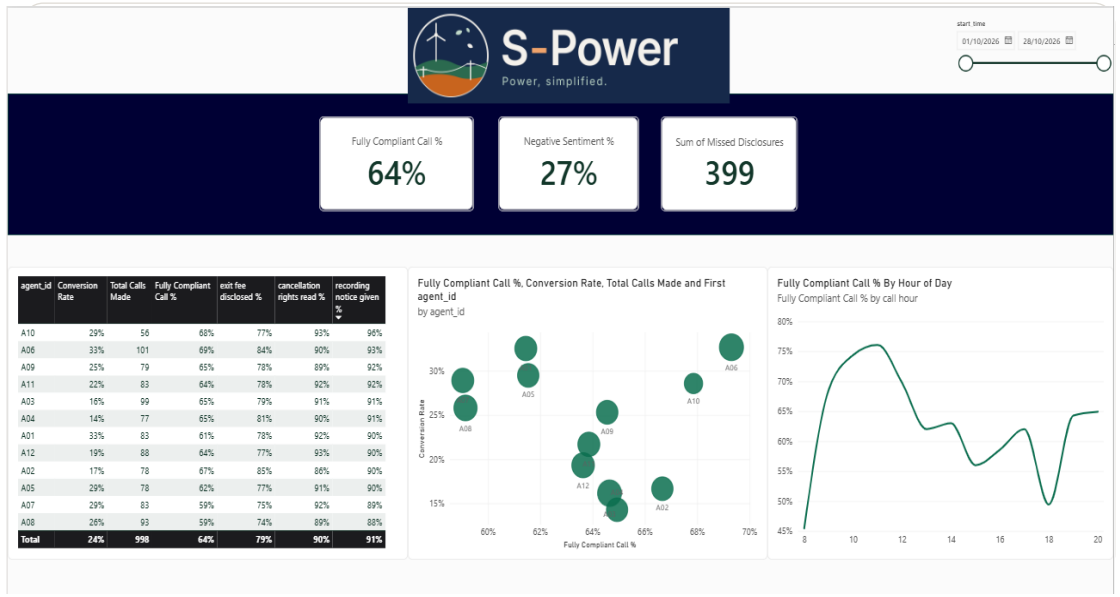
Power BI dashboard

Two report pages built on the BigQuery view. DAX measures calculate conversion rates, revenue per agent, compliance percentages and time-of-day patterns.

Page 1: Sales Performance



Page 2: Quality + Compliance



agent_id	Conversion Rate	Total Calls Made	Fully Compliant Call %	exit fee disclosed %	cancellation rights read %	recording notice given %
A10	29%	56	68%	77%	93%	96%
A06	33%	101	69%	84%	90%	93%
A09	25%	79	65%	78%	89%	92%
A11	22%	83	64%	78%	92%	92%
A03	16%	99	65%	79%	91%	91%
A04	14%	77	65%	81%	90%	91%
A01	33%	83	61%	78%	92%	90%
A12	19%	88	64%	77%	93%	90%
A02	17%	78	67%	85%	86%	90%
A05	29%	78	62%	77%	91%	90%
A07	29%	83	59%	75%	92%	89%
A08	26%	93	59%	74%	89%	88%
Total	24%	998	64%	79%	90%	91%

DAX Conversion %, revenue per agent, compliance rate, hourly patterns

Theme S-Power brand colours applied via custom JSON theme file



What comes next

1

Gold layer

Pre-aggregated tables for common queries: daily agent scorecards, weekly campaign summaries, rolling compliance trends.

2

Orchestration

Replace manual script runs with dbt for transformations and Airflow (or Cloud Composer) for scheduling and monitoring.

3

Live CCAI

Swap synthetic data for a real CCAI feed. The medallion structure and cleaning rules stay the same; only the source connector changes.



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Thank you

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